



DATA SCIENCE PROGRAM

CAPSTONE REPORT - SPRING 2026

The George Washington University

Data Science Program

Modeling Global Climate Change News: A Computational Analysis of Media Tone Across Countries

Capstone Report — Spring 2026

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May 10, 2026

Preface

While studying Data Science at The George Washington University, I completed an internship as a Data and Research Intern at Earthday.org, where I worked closely with climate-related datasets and communications. During that experience, I noticed that virtually all of the news and narratives I encountered before this were North American in origin. International and global perspectives on climate change were largely absent from my day-to-day exposure. This project grew directly out of that observation: I wanted to investigate whether measurable differences exist in how climate change is discussed in media across countries, and whether computational methods could reveal patterns in global climate narratives that are invisible when one only consumes regional news. This capstone gave me the opportunity to combine my interest in environmental issues with the machine learning and data analysis skills I developed throughout the Data Science program.

Acknowledgements

I would like to thank Prof. Sushovan Majhi for his guidance and feedback throughout this capstone project. His support in helping me think through both the technical and conceptual challenges of the work is greatly appreciated. I also thank the broader GW Data Science faculty for the foundation they provided over the course of the program. Finally, I am grateful to my colleagues and mentors at Earthday.org, whose work in climate advocacy first sparked my interest in the global dimensions of climate communication.

Abstract

Climate change is an increasingly concerning global crisis, yet its representation in the media is far from uniform. This project investigates how climate-related media tone varies across countries by applying a computational analysis to global news data drawn from the Global Database of Events, Language, and Tone (GDELT) in the year 2024. Climate-related articles were identified using Google BigQuery, and thematic features were extracted from the **V2Themes** field. After an extensive dimensionality reduction pipeline, including variance filtering, sparsity filtering, and correlation filtering, an initial feature space of approximately 4,000 variables was reduced to 98 thematic features spanning 118 countries. A multiple linear regression model was trained on an 80/20 train-test split to predict country-level average tone. The model achieved an $R^2 = 0.365$, indicating that thematic differences in news coverage explain a meaningful but partial portion of variation in media tone. Geospatial analysis revealed clear regional patterns: parts of Central Africa exhibited the most negative tone values, while Europe and parts of Asia tended toward more neutral or slightly positive tones. These findings demonstrate that climate narratives are shaped not only by geography but by the specific thematic emphasis of regional reporting. An F -test on the overall regression was statistically significant ($p < 0.05$), confirming that thematic features collectively explain a meaningful share of tone variance.

Keywords: climate change, media framing, GDELT, natural language processing, linear regression, global news analysis, media tone, computational social science

Contents

1	Introduction	6
1.1	Problem Statement and Motivation	6
1.2	Target Audience and Stakeholders	6
1.3	SMART Research Questions	6
1.4	Project Achievements and Roadmap	6
1.5	Scope and Non-Goals	7
2	Related Work and Background	8
2.1	Media Framing and Climate Change	8
2.2	Key Technical Concepts	8
3	Data: Sources, Description, and Preparation	9
3.1	Data Source and Provenance	9
3.2	Dataset Description	9
3.3	Licensing and Terms of Use	9
3.4	Data Quality and Issues	9
3.5	Data Preprocessing and Feature Engineering	10
3.6	Train/Test Split	10
4	Methods and Approach	11
4.1	Analytical Approach	11
4.2	Justification of Method	11
4.3	Baseline	11
4.4	Hyperparameters and Model Configuration	11
4.5	Evaluation Metric	11
4.6	Validation Strategy	11
5	Experimentation and Reproducibility	12
5.1	Experimental Setup	12
5.2	GitHub Repository	12
5.3	Environment	12
6	Results	13
6.1	Model Performance	13
6.2	Predicted vs. Actual Tone	13
6.3	Global Tone Distribution	14
6.4	Regional Theme Distribution	14
6.5	Addressing the SMART Research Questions	15
7	Discussion, Limitations, and Conclusion	16
7.1	Main Findings	16
7.2	Strengths Relative to Prior Work	16
7.3	Challenges Encountered	16
7.4	Limitations	16
7.5	Conclusion	16
7.6	Future Directions	17
8	Ethics, Data Use, and Responsible Practice	18
8.1	Ethical Considerations Throughout the Project	18
8.2	Bias and Fairness	18
8.3	Privacy and Consent	18
8.4	Potential Dual-Use and Downstream Harm	18
8.5	Concrete Mitigations Adopted	18

8.6 AI and Generative AI Use Disclosure	18
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1 Introduction

1.1 Problem Statement and Motivation

As climate crises continue to intensify, they have become a global issue shaped not only by environmental realities but also by how they are represented in the media. News coverage plays a critical role in shaping public understanding by highlighting certain aspects of climate change while downplaying others. These differences in reporting can influence how audiences perceive urgency, responsibility, and potential solutions, ultimately affecting public opinion and policy responses.

Importantly, the framing of climate issues is not uniform across countries. Regional, political, and cultural contexts influence which themes are emphasized, resulting in varied narratives around climate change worldwide. Some countries may focus on economic impacts, while others highlight environmental risks or policy debates. Understanding these differences is essential for identifying how global climate discourse is constructed and communicated.

1.2 Target Audience and Stakeholders

This project is primarily aimed at researchers in computational social science, climate communication scholars, policymakers, and environmental advocacy organizations who seek to understand global variation in media framing. It may also be of interest to journalists and media organizations aiming to assess the breadth of their climate coverage.

1.3 SMART Research Questions

This project is guided by the following SMART (Specific, Measurable, Achievable, Relevant, and Time-bound) research questions:

1. **RQ1:** To what extent does country-level average climate news tone, as measured by GDELT tone scores for 2024, vary across geographic regions?
2. **RQ2:** Which thematic features from GDELT’s `V2Themes` field are most strongly associated with variation in country-level average tone in 2024?
3. **RQ3:** Can a multiple linear regression model trained on 2024 thematic features predict country-level average climate news tone, and what R^2 does it achieve on a held-out test set?
4. **RQ4:** Are there identifiable regional patterns in climate tone across Africa, Asia, Europe, North America, and South America in 2024?

1.4 Project Achievements and Roadmap

This project successfully built an end-to-end pipeline for collecting, processing, and modeling global climate news data. A linear regression model with $R^2 = 0.365$ was trained and evaluated, and geospatial visualizations were produced to map tone variation across 118 countries. Clear regional differences were identified, with Central Africa showing the most negative average tone and parts of Europe and Asia showing more neutral or positive coverage.

The remainder of this report is structured as follows. Section 2 reviews related work on media framing, climate communication, and computational text analysis. Section 3 describes the data sources, collection process, and preprocessing pipeline. Section 4 details the modeling approach and evaluation strategy. Section 6 presents results including the regression model performance and geospatial tone distribution. Section 7 discusses the findings, limitations, and future directions. Section 8 addresses ethical considerations. Section 5 covers reproducibility and code availability.

1.5 Scope and Non-Goals

This project analyzes 2024 GDELT data only; longitudinal trend analysis across multiple years is outside scope. The project does not perform sentiment analysis at the article or sentence level, nor does it attempt causal inference about why tone differs. Translation and multilingual NLP are also out of scope — coverage is limited to whatever languages appear in GDELT.

2 Related Work and Background

2.1 Media Framing and Climate Change

A small number of countries are heavily featured in existing work on climate media framing and in nearly all such work there is reliance on manually coded samples of relatively small size. The most widely cited example of such work is Boykoff and Boykoff (1), who looked at US prestige press coverage of global warming from 1988 to 2002. They argue that the norm of journalistic balance led prominent outlets to give roughly equal coverage to the scientific consensus on global warming and to a handful of contrarians – a concept they termed ‘balance as bias’. The result was coverage which mis-stated the state of scientific consensus, and important background to the current project as it highlights the role of journalistic framing norms (not just underlying science) in climate reporting. The current project builds on this insight but applies it globally rather than in the US, and relies on an automated rather than a hand-coded analysis, across a much wider number of countries.

Painter (2) began to look at international comparisons of climate skepticism across six countries-Brazil, China, France, India, the U.K. And the U.S. The analysis covered 3,000 newspaper articles (two outlets per country) and indicated that climate skepticism was largely an Anglo-Saxon phenomenon centered in the U.S. And the U.K. Other countries like France and Brazil reported relatively little. That result motivated this project-if framing differed so drastically across six countries, what pattern was there in 118? The difference again lies in methodology. While Painter examined the hand-coded sample, this analysis used an aggregate of all articles published in countries across the GDELT project.

Nisbet (3) presents a relevant and useful framework in which to consider the media frames applied to climate issues. He argues for eight primary frames used when science communicators convey the science to the public: economic development, public accountability, scientific uncertainty, Pandora’s Box, risk/benefit, disaster, policy alternatives and political conflict frames. When comparing these to the thematic categories GDELT outputs, it appears to align well with the themes GDELT extracts which cover economic development, policy framing, and public safety/risk framing. The thematic indicators of GDELT therefore represent a good proxy for framing in this project. Feldman and Hart (4) looked at interactions between climate media framing and political predispositions in leading to selective news exposure, indicating how frame choice has downstream consequences for audience selection and response. This makes the case for the importance of differences in framing between countries.

Taken together, these studies agree that climate media framing is neither neutral nor uniform: it is shaped by journalistic norms, national context, and political orientation. Where they differ is in scope, prior studies examined a small number of countries manually, while this project scales to 118 countries computationally. Nisbet’s framing taxonomy provides the theoretical link between GDELT’s thematic tags and the concept of media framing, and the GDELT bias documentation frames the primary limitation of this approach.

2.2 Key Technical Concepts

GDELT (Global Database of Events, Language, and Tone) is a large-scale, open database that monitors news media worldwide. It assigns numerical tone scores to articles (positive values indicate more positive tone, negative values indicate more negative tone) and tags articles with thematic codes via the `V2Themes` field.

Multiple linear regression models the relationship between a continuous dependent variable (here, country-level average tone) and multiple predictor variables (here, thematic feature frequencies). The coefficient of determination R^2 measures the proportion of variance in the dependent variable explained by the model.

Dimensionality reduction via variance, sparsity, and correlation filtering removes uninformative or redundant features before modeling to reduce overfitting risk.

3 Data: Sources, Description, and Preparation

3.1 Data Source and Provenance

The primary data source is the Global Database of Events, Language, and Tone (GDELT) (5). Data was accessed via Google BigQuery (6) using a structured SQL query filtered for climate-related articles published in 2024. The GDELT dataset is freely available for research use under its open-access terms.

Source URL: <https://www.gdeltproject.org/>

Data accessed via: Google BigQuery public datasets

Date range: January 2024 – December 2024

Access date: Spring 2026

3.2 Dataset Description

Table 1: Dataset summary after preprocessing.

Attribute	Value
Unit of analysis	Country
Number of countries (rows)	118
Initial number of features	≈4,000
Final number of features	98
Time period	January – December 2024
Primary outcome variable	Average tone (continuous, GDELT scale)
Geographic scope	Global (118 countries)

The key variables are as follows. The outcome variable is **AvgTone**, a continuous numeric variable representing the mean GDELT tone score aggregated at the country level, where negative values indicate more negative sentiment and positive values indicate more positive sentiment. The predictor variables are thematic feature frequencies derived from the **V2Themes** field: each feature represents the proportion of climate-related articles from a given country that contain a specific GDELT theme tag (e.g., economic themes, environmental themes, policy themes).

3.3 Licensing and Terms of Use

GDELT data is publicly available and intended for research use. Google BigQuery provides access to GDELT as a public dataset under its standard terms of service. No personally identifiable information is present in this dataset; the unit of analysis is the country, not the individual. This project’s use of the data is consistent with the dataset’s open-access licensing.

3.4 Data Quality and Issues

Several data quality considerations were addressed during preprocessing. With respect to missing values, not all 195 UN-recognized countries had sufficient GDELT coverage in 2024 to be included. Of the 195 UN-recognized countries, 77 were dropped due to insufficient GDELT coverage in 2024, leaving 118 countries in the final dataset. Regarding sparsity, many of the approximately 4,000 initial thematic features were extremely sparse (i.e., present in very few countries), which risked inflating model complexity without adding predictive power. These were removed via sparsity filtering. Concerning coverage bias, GDELT is known to skew toward English-language sources Hong et al. (7), which may underrepresent countries where climate news is primarily published in other languages. This is acknowledged as a limitation.

3.5 Data Preprocessing and Feature Engineering

The preprocessing pipeline was implemented in Python using `pandas` and `scikit-learn` and proceeded in the following order.

First, **theme extraction** was performed: climate-related themes were extracted from the `V2Themes` field, which stores semicolon-separated theme tags. These were split into individual theme entries.

Second, **country-level aggregation** was carried out: theme frequencies were aggregated at the country level and transformed into a wide-format feature matrix, where rows represent countries and columns represent the frequency of each theme.

Third, **dimensionality reduction** was applied in three sequential steps: (a) variance filtering removed features with variance < 0.01 across countries; (b) sparsity filtering removed features present in fewer than 10% of countries; (c) correlation filtering removed features with pairwise Pearson correlation > 0.90 . This reduced the feature space from approximately 4,000 to 98 variables.

Fourth, **feature standardization** was performed: all features were standardized using `StandardScaler` (zero mean, unit variance) to ensure comparability across features with different raw scales.

Fifth, the **outcome variable** was computed: the average GDELT tone score was calculated for each country across all 2024 climate-related articles.

3.6 Train/Test Split

The dataset of 118 countries was split into an 80% training set ($n = 94$ countries) and a 20% test set ($n = 24$ countries) using a random split with a fixed random seed for reproducibility. No temporal split was required since the unit of analysis is country-level aggregates over the full 2024 period.

4 Methods and Approach

4.1 Analytical Approach

The primary analytical method is **multiple linear regression**, which models the country-level average GDELT tone score as a linear function of the 98 standardized thematic feature frequencies. Multiple linear regression was chosen because the outcome variable (average tone) is continuous, the predictors are numeric frequency features, and the goal is explanatory modeling — understanding which thematic features are associated with tone differences — rather than complex nonlinear prediction. The method is also transparent and interpretable, which is appropriate given that this is an exploratory analysis.

4.2 Justification of Method

Multiple linear regression is well-suited to this problem for several reasons. The research questions call for understanding the association between thematic features and tone (RQ2, RQ3), which linear regression directly addresses. The unit of analysis is country-level aggregates, which reduces the noise typical of individual-document analysis and makes a linear model more plausible. The model is interpretable, which supports the goal of identifying which thematic categories drive tone differences. A more complex model such as random forest or gradient boosting was considered but not implemented in this iteration; this is discussed as a future direction.

4.3 Baseline

As a sanity check and baseline, a **mean predictor** (predicting the training set mean tone for all countries) was used. The mean predictor achieves $R^2 = 0$ by definition. The linear regression model’s $R^2 = 0.365$ on the test set exceeds this baseline, confirming that the thematic features carry predictive signal beyond simply predicting the mean.

4.4 Hyperparameters and Model Configuration

The linear regression model was fit using `scikit-learn`’s `LinearRegression` with default settings (ordinary least squares). No regularization hyperparameters were tuned in the final model. During development, negative R^2 values were encountered on the test set with the initial feature set, indicating overfitting. This was resolved by applying the three-stage dimensionality reduction pipeline described in Section 3, which reduced the feature space from approximately 4,000 to 98 variables and eliminated the overfitting. Claude (Anthropic) and ChatGPT (OpenAI) were consulted during this debugging process; see Section 8 for full GAI disclosure.

4.5 Evaluation Metric

The primary evaluation metric is the **coefficient of determination** (R^2), which measures the proportion of variance in country-level average tone explained by the model. R^2 was chosen because the outcome is continuous and the goal is to assess how well thematic features explain tone variation. An R^2 closer to 1.0 indicates a better fit; $R^2 = 0$ corresponds to the mean baseline.

4.6 Validation Strategy

Model performance was evaluated on the held-out 20% test set. The dimensionality reduction steps (variance, sparsity, and correlation filtering) were designed and applied before the train/test split to avoid data leakage. Standardization was fit on the training set and applied to both training and test sets.

5 Experimentation and Reproducibility

5.1 Experimental Setup

The analysis was conducted in Python using a Jupyter Notebook environment. The key libraries used were `pandas` (data manipulation), `scikit-learn` (preprocessing, modeling, evaluation), and `matplotlib` / `geopandas` (visualization). Data was retrieved from GDELT via Google BigQuery using a structured SQL query filtering for climate-related articles in 2024.

The preprocessing pipeline, model training, and evaluation were all implemented in a single Jupyter Notebook. A fixed random seed was used for the train/test split to ensure reproducibility of results across runs.

5.2 GitHub Repository

The code for this project is available at the following GitHub repository:

GitHub Repository

<https://github.com/GW-datasci/-26-spring-Kelly-P>

The repository contains the Jupyter Notebook implementing the full pipeline from data loading through modeling and visualization. A `README.md` with setup instructions and a `requirements.txt` file capturing the Python environment are included.

5.3 Environment

The analysis was conducted using Python 3.x. Key package versions are listed in `requirements.txt` in the repository. The main dependencies are `pandas`, `scikit-learn`, `matplotlib`, and `google-cloud-bigquery`.

6 Results

6.1 Model Performance

The multiple linear regression model was evaluated on the held-out 20% test set (24 countries). The model achieved $R^2 = 0.365$, indicating that the 98 thematic features derived from GDELT’s `V2Themes` field explain approximately 36.5% of the variance in country-level average climate news tone. This result exceeds the mean-predictor baseline ($R^2 = 0$), confirming that thematic differences in news coverage carry meaningful predictive signal. An F -test on the overall regression was statistically significant ($p < 0.05$), confirming that the thematic features collectively explain a non-trivial share of tone variance beyond chance.

Table 2: Model performance summary on the held-out test set ($n = 24$ countries).

Model	R^2 (test)	Notes
Mean predictor (baseline)	0.000	Predicts training mean for all countries
Multiple linear regression	0.365	98 features, OLS, standardized inputs

6.2 Predicted vs. Actual Tone

The predicted versus actual tone plot (see Figure 2) shows that predicted tone values range from approximately -6 to 8 , while actual tone values range from approximately -5 to 3 . Data points follow an overall positive trend, consistent with a positive R^2 . Some dispersion is present, particularly at more extreme values. A small number of countries show notably higher predicted tone than observed, suggesting the model overestimates tone for some outlier countries. This dispersion is consistent with the moderate R^2 and reflects the inherent complexity of country-level tone, which is unlikely to be fully explained by thematic features alone.

□

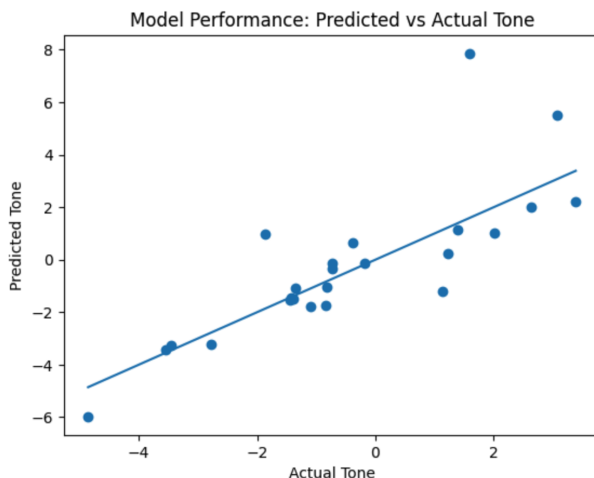


Figure 1: Linear Regression Tone Plot

Figure 2: Predicted versus actual country-level average tone for the held-out test set. Points follow an upward trend ($R^2 = 0.365$), with some dispersion at extreme values. The dashed line represents perfect prediction.

6.3 Global Tone Distribution

The global tone map (Figure 4) visualizes average climate news tone across 118 countries in 2024. Darker red colors represent more negative tone values and darker blue colors represent more positive tone values. Clear regional variation is observed: parts of Central Africa exhibit the most negative tone values, while regions in Europe and Asia tend toward more neutral to slightly positive tones. The United States and several countries in South America appear closer to neutral. This spatial pattern suggests that geographic and regional factors play a meaningful role in shaping climate media tone, above and beyond the thematic features captured in the regression model.

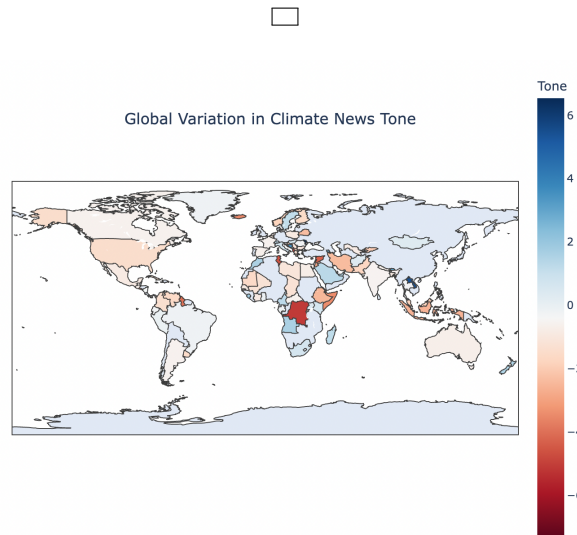


Figure 3: Comparison of Global Variation in Climate News Tone

Figure 4: Global variation in climate news tone across 118 countries, 2024. Darker red indicates more negative tone; darker blue indicates more positive tone. Regional variation is visible across Africa, Asia, Europe, North America, and South America.

6.4 Regional Theme Distribution

The distribution of climate themes varies across continents (Figure 6). Economic and policy-related topics appear consistently across all regions. Environmental themes such as forests and oceans show greater prominence in certain areas, particularly in South America. These differences indicate that climate coverage is not only varied in tone, but also in the specific topics emphasized, reflecting regional priorities and perspectives in climate discourse.

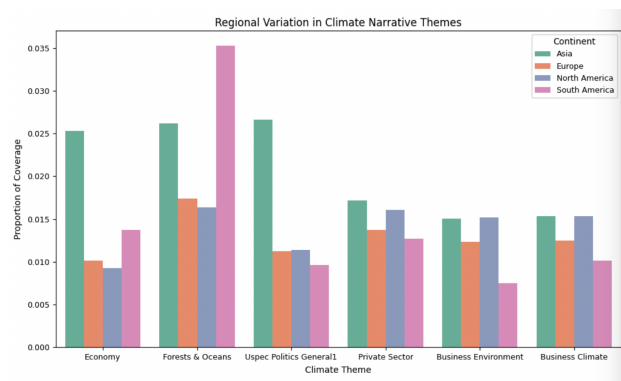


Figure 5: Describing Regional Variation in Climate Narrative Themes

Figure 6: Regional variation in climate narrative themes across continents (Asia, Europe, North America, South America), 2024 GDELT data. Theme categories shown on the x-axis; proportion of coverage on the y-axis.

6.5 Addressing the SMART Research Questions

RQ1 (regional variation in tone): Confirmed. The global tone map (fig. 4) reveals substantial variation, with Central Africa showing the most negative values and Europe and Asia showing more neutral to positive values.

RQ2 (thematic features associated with tone): Confirmed. The regression model identifies thematic features as statistically associated with tone. Economic and policy themes appear broadly, while environmental themes such as forests and oceans are prominent in specific regions.

RQ3 (predictability with linear regression): The model achieves $R^2 = 0.365$ on the held-out test set (table 2), indicating that thematic features explain a meaningful but partial proportion of tone variance.

RQ4 (identifiable regional patterns): Confirmed. Clear regional patterns are visible in both fig. 4 and fig. 6, with distinct differences across Africa, Asia, Europe, North America, and South America.

7 Discussion, Limitations, and Conclusion

7.1 Main Findings

This study examined how climate-related themes in global news coverage relate to differences in tone across countries in 2024. The results demonstrate that climate narratives vary significantly across geographic regions and are shaped by thematic emphasis in reporting. The global tone map revealed substantial variation, with Central Africa exhibiting the most negative tone values and Europe and parts of Asia trending toward more neutral or slightly positive tones. The regression analysis confirmed that thematic features are statistically associated with tone, with an $R^2 = 0.365$.

7.2 Strengths Relative to Prior Work

Compared to prior work such as Boykoff and Boykoff (1), this project offers several strengths. Unlike manual content analyses such as Painter (2), this project scales the analysis to 118 countries using automated feature extraction, enabling a truly global perspective. The use of GDELT provides a comprehensive and replicable data source, whereas many prior studies relied on bespoke article samples. The end-to-end Python pipeline is reproducible and extensible to future years or different query terms.

7.3 Challenges Encountered

The most significant technical challenge encountered during this project was model overfitting. The initial linear regression model with approximately 4,000 features produced a negative R^2 on the test set, indicating that the model was fitting noise in the training data rather than meaningful signal. This was resolved through the three-stage dimensionality reduction pipeline (variance, sparsity, and correlation filtering), which reduced the feature space to 98 variables and brought the test R^2 to 0.365.

7.4 Limitations

This project has several important limitations.

First, **linear regression is likely too simple a model** for this problem. With only 118 data points and 98 features, the model is still near the boundary of overfitting even after dimensionality reduction. Nonlinear models such as random forest or gradient boosting regression would likely capture more complex relationships between themes and tone. The $R^2 = 0.365$ reflects this limitation.

Second, **GDELT has known coverage biases**. The database skews heavily toward English-language sources Hong et al. (7), which may systematically underrepresent climate news from countries where reporting occurs primarily in other languages (e.g., Arabic, Mandarin, Portuguese). Countries with less English-language media may have their tone scores driven by a smaller and less representative sample of articles.

Third, **only 2024 data was used**, which limits the ability to identify longitudinal trends or assess whether the observed regional patterns are stable over time or reflect a particular moment in climate discourse (e.g., post-COP28 coverage).

Fourth, **the unit of analysis is the country**, which aggregates substantial within-country variation. Different media outlets, political leanings, and regional contexts within a country may produce very different tones that are averaged away in this analysis.

Possible mitigations for future work include incorporating multilingual data sources, expanding to multiple years, using regularized regression or tree-based models, and disaggregating the analysis to the outlet or region level within countries.

7.5 Conclusion

This project demonstrated that global climate news tone varies significantly across countries and regions and that thematic features derived from GDELT are associated with that variation. A multiple linear

regression model trained on 98 thematic features across 118 countries achieved $R^2 = 0.365$ on a held-out test set, explaining roughly one-third of the variance in country-level tone. Regional patterns in the geospatial visualization confirm that climate narratives are not uniform globally: Central Africa tends toward more negative coverage, while Europe and parts of Asia trend more neutral or positive. These findings provide a foundation for more nuanced computational analyses of global climate communication.

7.6 Future Directions

Building directly on the experience of this project, several concrete future directions are worth pursuing. First, replacing linear regression with a regularized model (ridge or lasso) or a tree-based ensemble (random forest, gradient boosting) would likely improve predictive performance while managing overfitting with a small sample size. Second, extending the analysis to multiple years (e.g., 2020–2024) would allow examination of whether regional tone patterns have shifted over time, particularly in response to major climate events or policy milestones. Third, incorporating multilingual GDELT data or complementary non-English news databases would reduce the English-language bias and provide a more representative global picture. Fourth, a finer-grained analysis at the outlet or sub-national level would reveal within-country variation currently obscured by country-level aggregation.

8 Ethics, Data Use, and Responsible Practice

8.1 Ethical Considerations Throughout the Project

Ethical considerations were addressed at each stage of this project. During data collection, GDELT was queried in accordance with its open-access terms of use. No data involving individual people’s personally identifiable information was collected or used; the unit of analysis is the country, not the individual, and all data consists of aggregate tone scores and theme frequencies derived from publicly available news articles. This eliminates direct privacy or identifiability concerns.

8.2 Bias and Fairness

The most significant bias concern in this project is the **English-language bias inherent in GDELT**. Because GDELT aggregates primarily English-language news sources, countries that publish climate-related news predominantly in other languages are likely underrepresented in the dataset, or represented by a smaller, less representative sample of articles. This means that tone scores for some countries — particularly in Africa, the Middle East, and parts of Asia — may reflect the tone of English-language international outlets covering those regions rather than domestic media. This is not a generic statement about data bias but a specific, documented property of the GDELT dataset Hong et al. (7). Results for countries with limited English-language media coverage should be interpreted with particular caution.

8.3 Privacy and Consent

This project does not involve personal data, survey data, or any information about identifiable individuals. All data is publicly available through GDELT and Google BigQuery, and its use is consistent with the dataset’s open-access terms. No re-identification risk is present, as the data is already aggregated to the country level.

8.4 Potential Dual-Use and Downstream Harm

The findings of this project do have the potential be misused. For example, a state actor or media organization could use knowledge of regional tone patterns to design targeted misinformation campaigns framed to exploit known regional sensitivities in climate coverage. Additionally, if the findings were misrepresented by claiming that negative tone in Central African climate coverage means that region is less concerned about climate change, which this could be used to downplay climate vulnerability in high-risk regions. These risks are acknowledged, and the report includes clear caveats about the limitations of GDELT coverage to reduce the likelihood of such misinterpretation.

8.5 Concrete Mitigations Adopted

Two concrete mitigations were adopted in this project. First, the analysis was conducted at the country level rather than the outlet or article level, reducing the risk that any single source’s idiosyncratic framing would drive the results. Second, the geographic bias of GDELT is explicitly disclosed throughout this report, particularly in the limitations section, so that readers understand the constraints of the data before drawing conclusions about any specific country or region.

8.6 AI and Generative AI Use Disclosure

Two generative AI tools were used during this project. **Claude (Anthropic, claude.ai)** and **ChatGPT (OpenAI, GPT-4)** were consulted during the debugging and model development phase of the project. Specifically, both tools were used to help diagnose and resolve a negative R^2 problem that arose when the initial linear regression model was applied to the full approximately 4,000-feature dataset. I used prompts: Why is my linear regression model producing negative R^2 on a small high-dimensional dataset. The AI tools suggested that the negative R^2 was likely caused by overfitting due to the high dimensionality of the feature space relative to the small sample size of 118 countries, and recommended dimensionality reduction strategies including variance filtering, sparsity filtering, and correlation filtering. I critically evaluated this advice,

implemented the suggested pipeline in the Jupyter Notebook independently, and verified that the approach successfully resolved the overfitting issue (raising the test R^2 from negative to 0.365). Risks considered included the possibility of hallucinated function names or incorrect **scikit-learn** API details; all code suggestions were verified against the official **scikit-learn** documentation before use.

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